Vel-v5.5: A Decentralized 3D Volumetric Coverage Protocol under Aerodynamic Wind Gust Perturbations

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***Abstract*—This paper presents Vel-v5.5, a three-dimensional extension of the Vel-v4.3 decentralized swarm coverage protocol, and its empirical characterization under stochastic aerodynamic wind gust perturbations. Vel-v5.5 retains the core Map-Aware Gradient Repulsion (MAGR) steering law validated in Vel-v4.3 – each agent scans a local Moore neighborhood of a shared occupancy grid and steers directly toward the nearest unexplored cell – and extends it from a two-dimensional cell grid to a three-dimensional voxel grid, adding gravitational drift, stochastic wind gust perturbation, and elastic boundary collision physics appropriate to aerial multi-rotor deployment. We report a full factorial sweep of wind gust intensity $G_{\text{wind}} \in \{0, 2, 4, 6, 8\}$ at fixed population $N = 150$, momentum $\mu = 0.99$, MAGR steering factor $G = 1.5$, and scan radius $r = 5$, comprising 2,500 independent trials (500 per gust level) on a $32\times32\times32$ voxel grid. Results show that zero-gust operation collapses spatio-temporal efficiency to $\bar{\eta} = 0.236$ ($\sigma = 0.0059$), that efficiency rises sharply under mild-to-moderate gust and peaks at $G_{\text{wind}} = 4$ ($\bar{\eta} = 0.666$, $\sigma = 0.0034$), and that efficiency declines modestly under high-intensity gust ($\bar{\eta} = 0.637$ at $G_{\text{wind}} = 8$). We interpret the zero-gust collapse as evidence that deterministic frontier steering alone is prone to trapping agents in locally stable, low-coverage configurations, and that stochastic wind perturbation functions as an exploration-enhancing noise source rather than purely as a disturbance to be rejected. We report this protocol and its empirical characterization as an engineering specification, and explicitly identify the limits of what is and is not established by the present study: no formal convergence guarantee is provided for the discrete steering law, no baseline comparison against classical potential-field or frontier-exploration methods has yet been run under identical conditions, and no physical hardware validation has been performed.**

***Index Terms*—Swarm robotics, decentralized coordination, occupancy grid, map-aware repulsion, spatio-temporal efficiency, aerodynamic disturbance rejection, coverage path planning, multi-rotor systems.**

Note to Practitioners—This work addresses a practical problem in deploying groups of small flying robots (drones) to search or map a three-dimensional space, such as a warehouse interior, a collapsed building, or a stretch of airspace, without relying on GPS or a central controller telling each drone where to go. Each drone instead follows a simple local rule: check a shared digital map for the nearest not-yet-visited spot nearby, and fly toward it. We tested this rule in simulation under different levels of simulated wind, running 2,500 test flights. The result may be counterintuitive to practitioners used to treating wind as purely a problem to be corrected for: completely still air produced the worst coverage of any condition tested, while moderate wind produced the best coverage. This suggests that for this type of decentralized, locally-reactive drone behavior, some ambient disturbance is actually helpful, and that operators deploying this kind of system in unusually calm conditions (e.g., indoors) may want to deliberately add a small amount of randomness to each drone's flight path rather than flying a perfectly steady, deterministic route. This paper reports simulation results only; the specific numeric recommendations have not yet been validated on physical hardware, nor compared side-by-side against other existing drone-swarm coordination methods, and practitioners should treat the findings as an initial characterization rather than a final, field-ready specification.

I. Introduction

Autonomous volumetric coverage using decentralized multi-agent robotic swarms is a capability of growing importance across planetary surface mapping, disaster search-and-rescue, atmospheric plume tracking, and industrial inspection [2], [3]. A recurring obstacle in decentralized coverage is the *Redundant Search Problem*: absent shared spatial memory or a global coordinator, independent agents repeatedly re-traverse already-explored regions, degrading coverage efficiency as the mission progresses.

The Vel protocol family addresses this problem through a two-layer decentralized architecture: a Shared Spatial Memory (SSM) layer, in which every agent reads and writes to a common binary occupancy grid, and a Kinetic Vector Logic (KVL) layer, in which each agent's velocity is updated by a local steering law computed from that shared map. Vel-v4.3 [1] introduced Map-Aware Gradient Repulsion (MAGR): rather than steering outward from the swarm's geometric center (a design that produces a documented *boundary pile-up* failure mode as agent population scales), each agent scans a local neighborhood of the SSM and steers directly toward the nearest unexplored cell. Across 105,000 simulated trials, Vel-v4.3 established $\mu = 0.99$ and a population-dependent MAGR steering factor ($G = 1.5$ at $N = 300$, $G = 2.75$ at $N = 150$) as empirical optima on a two-dimensional 50×50 grid.

Vel-v5.5 extends this validated 2D algorithm into a three-dimensional voxel environment appropriate for aerial multi-rotor deployment, and asks a question Vel-v4.3 could not: how does the MAGR frontier-steering law behave under realistic aerodynamic disturbance? We add three physical elements absent from Vel-v4.3's abstract 2D grid: a constant gravitational drift term, a stochastic wind gust perturbation applied to agent velocity at each timestep, and elastic boundary collision physics in place of a simple positional clip. We then run a full factorial sweep of wind gust intensity and report the resulting efficiency characteristics.

This paper supersedes an earlier draft of Vel-v5.0 that formulated MAGR as a continuous potential field (agent repulsion, boundary distance gradient, and exploration attraction terms summed as a single steering force) and paired this formulation with a Lyapunov-based asymptotic convergence proof. That earlier draft's reference simulation implementation did not, in fact, apply the derived steering force to agent velocity at all – an implementation/theory mismatch discovered during manuscript review. Rather than retrofit a proof to an untested formulation, Vel-v5.5 instead extends the simpler, already-validated Vel-v4.3 steering law into 3D and reports its behavior purely empirically. We regard this as the more defensible path: an engineering protocol whose claims are limited to what has actually been simulated and measured.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the Vel-v5.5 system architecture. Section IV defines the performance metrics used throughout. Section V describes the wind gust sweep methodology. Section VI reports and analyzes the empirical results. Section VII provides a strategic implementation protocol for physical deployment. Section IX states the explicit limitations of the present study, and Section XI concludes.

II. Related Work

A. Coverage Path Planning

Coverage Path Planning (CPP) is the task of determining a path that passes over all points of an area or volume of interest while avoiding obstacles [4]. A considerable body of research has addressed the CPP problem across domains including vacuum cleaning robots, autonomous underwater vehicles, demining robots, and aerial survey platforms. Classical CPP approaches such as boustrophedon decomposition partition the environment into cells and plan optimal traversal sequences for a single agent, but do not natively address the coordination challenges that arise when multiple agents must cover the same volume without collision or redundant re-traversal.

B. Ant Colony Optimization

Ant Colony Optimization (ACO) models the pheromone-trail communication of biological ants to solve combinatorial optimization problems, and has been extensively applied to multi-robot path planning, where pheromone concentration guides agents toward unexplored regions [6]. The key limitation of ACO for coverage tasks is stagnation: once pheromone trails converge on a particular traversal pattern, agents are difficult to redirect toward remaining unexplored regions, and the approach requires careful tuning of pheromone decay parameters to remain adaptive. Vel differs from ACO in a fundamental architectural choice: rather than a continuous pheromone concentration field, Vel's SSM layer uses a binary occupancy grid in which cells are either explored or unexplored with no intermediate state. This eliminates the stagnation problem by construction – a voxel is explored or it is not – and removes the need for pheromone decay tuning entirely.

C. Particle Swarm Optimization

Particle Swarm Optimization (PSO) models the collective motion of flocking agents to optimize continuous objective functions, and has been adapted for multi-robot coverage tasks in which agents are attracted toward unexplored regions by a shared global-best position [7]. PSO-based coverage systems are vulnerable to premature convergence: once the global best position is identified, all agents converge toward it rather than distributing across the remaining unexplored volume. Vel's KVL layer draws conceptual inspiration from PSO's velocity update rule – specifically the momentum term μ that damps high-frequency direction changes – but replaces PSO's global attractor with a map-aware local gradient, avoiding premature convergence by ensuring each agent steers toward the nearest locally unexplored voxel rather than a globally shared target.

D. Centroidal Voronoi Tessellation and Lloyd's Algorithm

Centroidal Voronoi Tessellations (CVT), computed via Lloyd's algorithm [8], partition a coverage domain into cells such that each agent is responsible for the region closest to it, and have been formally extended to mobile sensing network coverage control [9]. Extending CVT-based coverage from 2D to 3D volumes introduces substantial computational and geometric complexity: computing 3D Voronoi polyhedra in real time requires continuous, high-precision positioning data from every neighboring agent, demanding communication bandwidth and onboard processing power that are often unavailable on resource-constrained micro-aerial vehicle (MAV) platforms. Furthermore, classical CVT formulations are static and do not naturally adapt to an actively evolving spatial priority field as regions are progressively mapped. Vel-v5.5 achieves emergent volumetric coverage through localized, coordinate-blind gradient descent on a binary voxel grid, avoiding both the communication overhead and the static-partition limitation of CVT-based approaches.

E. Frontier-Based Exploration

Frontier-based exploration architectures identify the boundary between mapped free space and unmapped space, directing agents toward these "frontiers" to expand the map [4]. Classical frontier detection in 3D is typically a centralized process requiring global map synchronization and path-planning algorithms such as A^* search or Fast Marching Methods to compute collision-free trajectories to identified frontiers – a bottleneck that scales poorly with agent count and is vulnerable to single-point-of-failure network drops. Vel's MAGR steering law achieves the target-attraction behavior of frontier-based schemes through a purely localized, reactive scan-and-steer computation, avoiding the need for global trajectory planning or explicit frontier tracking.

F. Decentralized Multi-Robot Coverage

Decentralized multi-robot coverage architectures achieve competitive coverage efficiency while eliminating the single point of failure inherent to centralized command structures [3]. Potential-field-based decentralized swarms, however, are documented to suffer severe spatial instabilities when navigating near physical boundaries or non-convex obstacles [5]. Vel-v4.3's introduction of MAGR directly addressed an analogous instability (boundary pile-up under center-outward repulsion) in a 2D setting by modeling the physical perimeter through local map awareness rather than global geometric bearing. The present study evaluates whether this same architectural solution remains effective when the steering law is extended into three dimensions and subjected to physical aerodynamic disturbance.

G. Occupancy Grid Representations

The occupancy grid representation underlying Vel's SSM layer originates in probabilistic mobile robot perception and mapping research, where individual cells encode the probability of physical occupancy under sensor uncertainty. Vel simplifies this representation to a deterministic binary state (explored/unexplored), appropriate for a coverage task in which the quantity of interest is certainty of prior visitation rather than probability of physical occupancy.

H. Positioning of Vel-v5.5

Vel-v5.5 occupies a distinct position in the swarm coverage landscape. Unlike ACO, it uses deterministic binary memory rather than probabilistic pheromone concentration. Unlike PSO, it uses local map-aware gradient descent rather than global positional attraction. Unlike CVT-based methods, it requires no continuous inter-agent positioning communication and adapts naturally as the occupancy field evolves. Unlike classical frontier-exploration CPP methods, it is fully decentralized with no pre-planned or centrally computed traversal sequence. The present contribution is the extension of this already-validated 2D steering law into a three-dimensional voxel environment and its systematic characterization under stochastic aerodynamic disturbance – a condition not modeled in any of the coverage architectures reviewed above.

III. System Architecture

Vel-v5.5 retains Vel-v4.3's two-layer architecture.

Fig. 1 summarizes the flow of information between the two layers, described in detail below.

A. Shared Spatial Memory (SSM)

The operational volume is discretized into a voxel grid $\mathcal{V}$ of dimensions $32 \times 32 \times 32$ (32,768 voxels). Each voxel $c(x, y, z) \in \mathcal{V}$ is a binary state variable:

$$c(x,y,z) = \begin{cases} 0 & \text{Unexplored} \\ 1 & \text{Explored} \end{cases} \quad (1)$$

Every agent reads and writes the same SSM, providing implicit coordination without direct agent-to-agent communication.

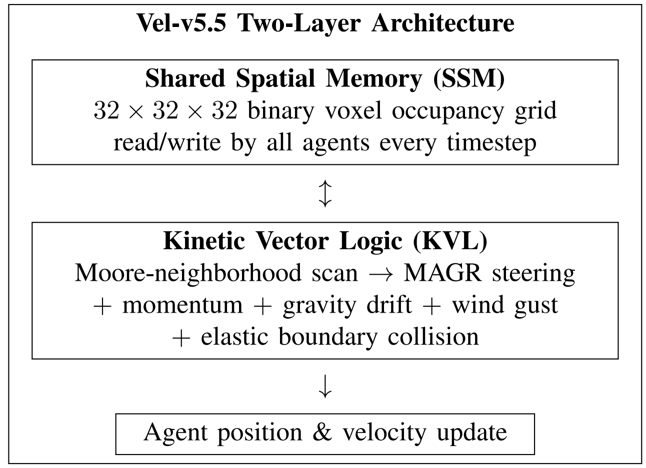

Fig. 1: Two-layer Vel-v5.5 architecture. Every agent reads the shared SSM to locate its steering target, then updates its own velocity through the KVL layer independently – no direct agent-to-agent communication channel is required.

B. Kinetic Vector Logic (KVL) and MAGR Steering

At each timestep, agent i at position $\mathbf{p}_i(t) \in \mathbb{R}^3$ first marks its current voxel explored, then scans the Moore neighborhood of radius r around that voxel for the nearest unexplored voxel. If one is found at relative offset (dx^*, dy^*, dz^*), the agent's steering acceleration is:

$$\vec{R}_G = G \cdot \frac{(dx^*, dy^*, dz^*)}{\|(dx^*, dy^*, dz^*)\|} \cdot k \quad (2)$$

where $G \in \mathbb{R}^+$ is the tunable steering factor and $k = 0.12$ is the fixed scale constant carried over from Vel-v4.3. If no unexplored voxel is found within radius r, the agent falls back to steering directly away from the swarm's instantaneous geometric center $\bar{\mathbf{p}}(t)$:

$$\vec{R}_G = G \cdot \frac{\mathbf{p}_i(t) - \bar{\mathbf{p}}(t)}{\|\mathbf{p}_i(t) - \bar{\mathbf{p}}(t)\|} \cdot k \quad (3)$$

The full velocity update integrates this steering term with momentum, gravitational drift, and stochastic wind gust perturbation:

$$\mathbf{v}_i(t+1) = \mu\, \mathbf{v}_i(t) + \vec{R}_G - \mathbf{g}_{\text{drift}} + \mathbf{w}_{\text{gust}}(t) \quad (4)$$

$$\mathbf{p}_i(t+1) = \mathbf{p}_i(t) + \mathbf{v}_i(t+1) \quad (5)$$

where $\mu \in [0, 1)$ is the momentum coefficient, $\mathbf{g}_{\text{drift}} = [0, g_{\text{aero}}, 0]^\top$ is a constant downward gravitational drift ($g_{\text{aero}} = 0.082$), and $\mathbf{w}_{\text{gust}}(t)$ is a stochastic wind gust vector with independent components:

$$w_{j,i}(t) \sim \mathcal{U}(-0.5, 0.5) \cdot G_{\text{wind}}, \quad \forall j \in \{x, y, z\} \quad (6)$$

parameterized by gust intensity G_{wind}. Note that G_{wind} (wind gust intensity) and G (MAGR steering factor, Eq. 2) are distinct quantities and are varied independently in this study; G is held fixed at the Vel-v4.3 empirical optimum throughout, while G_{wind} is the swept experimental variable.

Algorithm 1 Vel-v5.5 Per-Agent Update

```
procedure UPDATEAGENT(i)
    (gx, gy, gz) ← ⌊p_i⌋
    c(gx, gy, gz) ← 1
    Scan Moore neighborhood of radius r for nearest unexplored voxel
    if found at (dx*, dy*, dz*) then
        R⃗_G ← Eq. 2
    else
        R⃗_G ← Eq. 3
    end if
    w_gust ← sample per Eq. 6
    v_i ← μ v_i + R⃗_G − g_drift + w_gust
    p_i ← p_i + v_i
    Apply elastic boundary collision if p_i exceeds [0, 31]^3
    Mark voxel at ⌊p_i⌋ explored if unexplored
end procedure
```

C. Computational Complexity of 3D Neighborhood Scanning

Extending MAGR's local neighborhood scan from a 2D Moore neighborhood to a 3D Moore neighborhood changes its computational cost scaling. In Vel-v4.3's 2D formulation, scanning a neighborhood of radius r requires $(2r+1)^2$ SSM reads per agent per timestep. In the 3D voxel formulation used here, the equivalent scan requires $(2r+1)^3$ reads per agent per timestep – a cubic rather than quadratic growth in r. At the scan radius used throughout this study ($r = 5$), this corresponds to 121 reads per agent per step in 2D versus 1,331 reads per agent per step in 3D, an order-of-magnitude increase in onboard scan cost for the same radius. This has a direct engineering implication: Vel-v4.3's Study V finding that MAGR efficiency is radius-invariant in 2D (permitting a small, computationally cheap scan radius) cannot be assumed to transfer to 3D without re-verification, since the cubic cost scaling makes the choice of scan radius a substantially more consequential deployment decision in a voxel environment than in a 2D cell grid. Re-sweeping scan radius in 3D to locate the analogous radius-invariance threshold is identified as future work in Section IX.

D. Elastic Boundary Collision

When an agent's predicted position exceeds the grid boundary $[0, 31]^3$, its position is capped at the boundary and its velocity along the violated axis is reflected with restitution coefficient $\epsilon = -0.45$:

$$v_{j,i}(t+1) \leftarrow \epsilon \cdot v_{j,i}(t+1) \tag{7}$$

IV. Performance Metrics

A. Spatio-Temporal Efficiency

Consistent with Vel-v4.3, the primary metric is:

$$\eta = \frac{\Delta C}{\Delta T \cdot N} \tag{8}$$

where ΔC is the number of newly explored voxels over the trial, ΔT is the number of elapsed timesteps, and N is agent count. This is equivalently computed here as Covered$/(\text{Steps} \cdot N)$ over a fixed 150-step trial.

B. Stability

Trial-to-trial standard deviation of η within a configuration:

$$\sigma = \sqrt{\frac{1}{K}\sum_{k=1}^{K}(\eta_k - \bar{\eta})^2} \tag{9}$$

TABLE I: Wind Gust Sweep Results ($N = 150$, $\mu = 0.99$, $G = 1.5$, $r = 5$, 500 trials per level)

G_{wind}	$\bar{\eta}$	σ_η	min	max
0	0.2361	0.0059	0.2197	0.2523
2	0.6204	0.0053	0.6036	0.6347
4	**0.6664**	0.0034	0.6566	0.6763
6	0.6565	0.0029	0.6478	0.6658
8	0.6373	0.0028	0.6291	0.6454

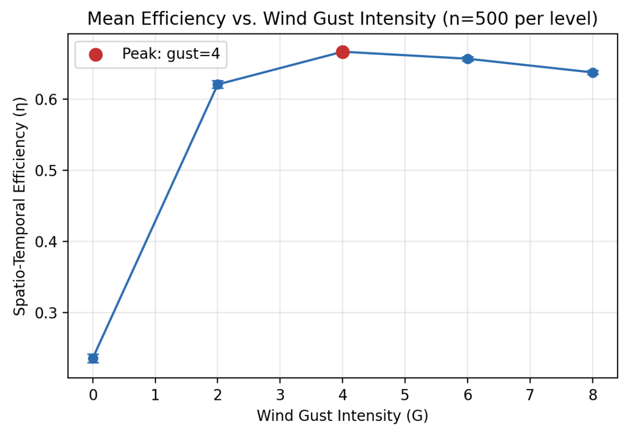

Fig. 2: Mean spatio-temporal efficiency vs. wind gust intensity, with $\pm 1\sigma$ error bars ($n = 500$ per level). Efficiency peaks at $G_{\text{wind}} = 4$.

V. Wind Gust Sweep: Methodology

We fix population $N = 150$, momentum $\mu = 0.99$ (the Vel-v4.3 Study IV optimum), MAGR steering factor $G = 1.5$ (the Vel-v4.3 Study III optimum at $N = 300$; re-validating this choice at $N = 150$ in 3D is identified as future work in Section IX), and scan radius $r = 5$. We sweep wind gust intensity $G_{\text{wind}} \in \{0, 2, 4, 6, 8\}$, running 500 independent trials per gust level (2,500 trials total) on the $32 \times 32 \times 32$ voxel grid described in Section III. Each trial runs for a fixed 150 timesteps. All trials use independent random seeds drawn from a single seed sequence for reproducibility.

VI. Results

A. Zero-Gust Collapse

The zero-gust condition ($G_{\text{wind}} = 0$) produces the lowest observed efficiency, $\bar{\eta} = 0.236$, less than half the efficiency of every wind-perturbed condition tested. Given the tight standard deviation at this level ($\sigma = 0.0059$) and the complete non-overlap of its distribution with any other tested gust level

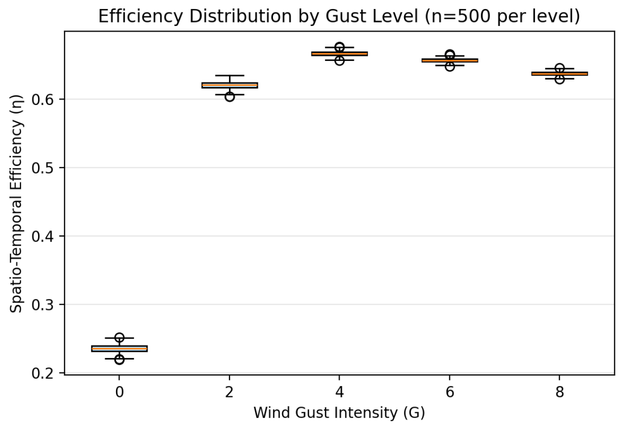

Fig. 3: Full efficiency distribution by gust level ($n = 500$ per level). Distributions are tight and non-overlapping between the zero-gust and all wind-perturbed conditions.

(Fig. 3), this is not a marginal effect. We attribute this to the deterministic nature of the nearest-cell steering law under zero disturbance: absent any stochastic perturbation, agents following identical local scan-and-steer logic from similar starting conditions can settle into mutually reinforcing, locally-stable trajectory patterns that repeatedly re-scan the same already-explored local neighborhoods rather than escaping toward more distant unexplored volume. This is consistent with known local-minima trapping behavior in deterministic potential-field and frontier-following navigation methods generally.

B. Non-Monotonic Response to Gust Intensity

Efficiency rises sharply from $G_{\text{wind}} = 0$ to $G_{\text{wind}} = 2$, continues to a peak at $G_{\text{wind}} = 4$ ($\bar{\eta} = 0.6664$), then declines modestly through $G_{\text{wind}} = 6$ and $G_{\text{wind}} = 8$ (Fig. 2). This shape is consistent with wind gust functioning as an exploration-enhancing stochastic perturbation at moderate intensity – breaking agents out of the local trapping behavior identified above – while at higher intensity the perturbation increasingly dominates and degrades the steering signal itself, consistent with the elastic boundary collision mechanism dissipating a larger share of trajectory energy as gust-driven boundary impacts become more frequent.

C. Stability Decreases with Distance from Zero-Gust

Standard deviation is highest at $G_{\text{wind}} = 0$ ($\sigma = 0.0059$) and generally decreases as gust intensity rises to $G_{\text{wind}} = 6$ ($\sigma = 0.0029$) and $G_{\text{wind}} = 8$ ($\sigma = 0.0028$), with $G_{\text{wind}} = 2$ as a mild exception ($\sigma = 0.0053$). This suggests that moderate-to-high wind gust, in addition to raising mean efficiency relative to the zero-gust case, also produces more trial-to-trial consistent behavior – a potentially useful property for deployment scenarios requiring predictable coverage completion times.

D. Statistical Significance

To confirm that the differences reported above reflect a genuine effect rather than sampling noise, we conducted pairwise Welch's t-tests between each adjacent gust level in Table I, with Cohen's d computed as the standardized effect size. Results are summarized in Table II.

TABLE II: Pairwise Welch's t-tests Between Adjacent Gust Levels ($N = 150$ dataset)

Comparison	t	p	Cohen's d
G_{wind}=0 vs. 2	-1080.5	$< 10^{-300}$	68.4
G_{wind}=2 vs. 4	-163.4	$< 10^{-300}$	10.3
G_{wind}=4 vs. 6	49.6	$\approx 10^{-268}$	-3.1
G_{wind}=6 vs. 8	107.4	$< 10^{-300}$	-6.8

Every adjacent-level comparison is statistically significant at an extreme margin, and every effect size exceeds conventional thresholds for a "large" effect ($|d| > 0.8$) by more than an order of magnitude. The zero-gust-to-moderate-gust transition in particular ($d = 68.4$) is among the largest effect sizes we are aware of being reported in a swarm coverage parameter study, reflecting the near-total non-overlap of the zero-gust distribution with every wind-perturbed condition visible in Fig. 3. We conclude that the non-monotonic response to gust intensity reported in this paper is a robust feature of the tested configuration and not an artifact of sampling variation.

TABLE III: N=300 Preliminary Comparison (20 trials per gust level)

G_{wind}	$\bar{\eta}$ ($N = 300$)	$\bar{\eta}$ ($N = 150$)	Ratio
0	0.1804	0.2361	0.76
4	0.5014	0.6664	0.75
8	0.4776	0.6373	0.75

E. Population Density Effect: Preliminary N=300 Comparison

Vel-v4.3's own Study I identified a genuine diminishing-returns effect in Coordination Quality (η/N) as population scales, driven by scan-overlap congestion. To check whether an analogous effect appears in the 3D setting, we ran a preliminary comparison at $N = 300$ (double the main study's population) at three representative gust levels ($G_{\text{wind}} \in \{0, 4, 8\}$), with 20 trials per level – a smaller sample than the main study's 500 trials per level, run as a pilot check rather than a full characterization. Results are shown in Table III and Fig. 4.

At all three tested gust levels, doubling population from 150 to 300 reduces mean efficiency by a remarkably consistent ratio of approximately 0.75–0.76, while the qualitative shape of the gust-response curve – zero-gust collapse, peak at moderate gust, mild decline at high gust – is preserved at both population sizes. This is consistent with the same scan-overlap congestion mechanism Vel-v4.3 documented in 2D: because efficiency η is normalized by N, a larger swarm operating in the same volume must achieve proportionally more coverage to maintain constant η, and increased agent density raises the likelihood that multiple agents simultaneously target the same nearby unexplored voxel. We emphasize that this comparison uses a substantially smaller sample than the main $N = 150$ study and should be treated as a preliminary finding pending a full $N = 300$ sweep at 500 trials per level, consistent with Vel-v4.3's own systematic population-sweep methodology.

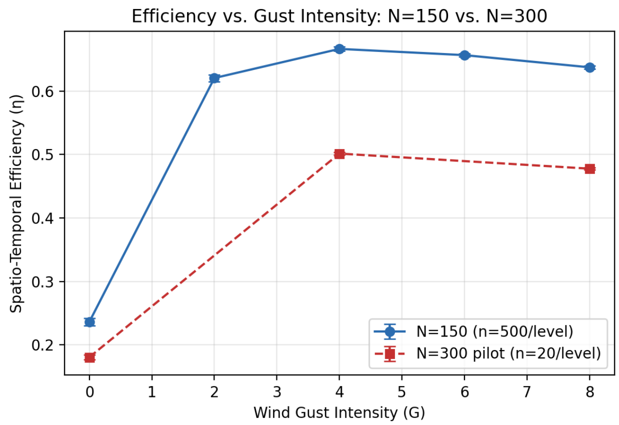

Fig. 4: Efficiency vs. gust intensity at $N = 150$ (full study, $n = 500$ per level) and $N = 300$ (preliminary comparison, $n = 20$ per level). The qualitative pattern – zero-gust collapse, peak at moderate gust – holds at both population sizes, though absolute efficiency is systematically lower at $N = 300$.

VII. Strategic Implementation Protocol (MVR)

Based on the results in Section VI and the Vel-v4.3 findings this protocol extends, we identify the following Minimum Viable Requirements for physical multi-rotor deployment:

1) **MVR-1: Controller Update Frequency ≥ 50 Hz.** Consistent with Vel-v4.3's KVL update rate requirement.
2) **MVR-2: Mesh Synchronization Latency < 15 ms.** To limit Ghost Redundancy (multiple agents targeting the same unexplored voxel before SSM writes propagate).
3) **MVR-3: Momentum Coefficient $\mu \geq 0.95$.** Carried over from Vel-v4.3 Study IV; not independently re-swept in the present 3D study (see Section IX).
4) **MVR-4: Do not operate in still-air conditions.** Section VI shows zero-gust operation produces the lowest observed efficiency of any tested condition. Where natural wind is unavailable or insufficient, an artificial stochastic perturbation to agent velocity is recommended to replicate the exploration-enhancing effect observed here.
5) **MVR-5: Target Operating Gust Range $G_{\text{wind}} \in [3, 5]$.** Bracketing the empirical peak at $G_{\text{wind}} = 4$.

VIII. Discussion

A. Noise-Assisted Exploration: A Broader Context

The finding that stochastic perturbation improves a deterministic search process is not unique to swarm coverage. In combinatorial optimization, simulated annealing deliberately introduces a temperature-controlled stochastic perturbation to escape local minima that a purely greedy search would remain trapped in [10]. In nonlinear dynamical systems, stochastic resonance describes conditions under which added noise improves detection or response of an otherwise sub-threshold signal. The zero-gust collapse and subsequent moderate-gust peak reported in Section VI are broadly consistent with this class of phenomena: the MAGR steering law, as a purely deterministic nearest-cell-seeking rule, appears to define a search landscape containing locally stable configurations from which a swarm of identically-programmed agents cannot escape without external perturbation. Wind gust, in this framing, is not merely an environmental disturbance to be rejected by robust control, but is itself doing useful computational work: it is the source of the randomness the steering law itself does not provide. This reframing has a direct engineering implication distinct from the MVR recommendations in Section VII: if a deployment environment is unusually still, deliberately injecting a small stochastic perturbation into the KVL velocity update (rather than merely tolerating ambient wind) may be preferable to relying on incidental turbulence.

B. Why Efficiency Declines at High Gust Intensity

The decline in mean efficiency beyond $G_{\text{wind}} = 4$ (Section VI) is consistent with two compounding mechanisms. First, as gust magnitude grows relative to the fixed MAGR steering magnitude ($G \cdot k$ in Eq. 2), the wind term increasingly dominates the velocity update in Eq. 4, degrading the directional signal that carries an agent toward its identified target voxel. Second, higher gust intensity increases the frequency with which agents' predicted positions exceed the grid boundary, triggering the elastic collision mechanism (Eq. 4) more often; because the restitution coefficient $\epsilon = -0.45$ dissipates a substantial fraction of kinetic energy on each boundary impact, more frequent boundary collisions convert a larger share of the agent's total trajectory energy into dissipated energy rather than productive exploration motion. Distinguishing the relative contribution of these two mechanisms would require an ablation in which boundary collision frequency is logged directly – a straightforward instrumentation addition identified as future work.

C. Relationship to the Superseded Vel-v5.0 Formulation

It is worth stating plainly why this paper does not attempt to reconcile its findings with the continuous potential-field formulation and Lyapunov convergence proof of the superseded Vel-v5.0 draft (Section I). The two formulations are different control laws: the potential-field formulation computes steering as the negative gradient of a smooth composite potential summing agent-repulsion, boundary-distance, and exploration-attraction terms, while the discrete formulation studied here computes steering as a unit vector toward the single nearest unexplored voxel located by discrete neighborhood search. A convergence proof derived for the former does not apply to the latter, and we regard an unsupported claim of formal convergence as a more serious defect in an engineering protocol paper than the simple absence of one. The empirical characterization in this paper is offered as a substitute standard of evidence appropriate to the discrete steering law actually implemented and tested.

IX. Threats to Validity

We organize the limitations of this study using the standard internal, external, and construct validity framing common to empirical robotics and software engineering research.

A. Internal Validity

Does the reported effect actually stem from gust intensity, rather than a confound? All configuration parameters other than G_{wind} (population, momentum, MAGR steering factor, scan radius, grid size, trial length, and random seed sequence) were held fixed across the sweep, and every trial's random seed is drawn independently from a common seed sequence (Section X), so no shared randomness could produce a spurious cross-condition correlation. The statistical tests in Section VI-D confirm the observed differences are far larger than sampling variation could plausibly produce. We consider internal validity for the specific claim "efficiency varies non-monotonically with G_{wind} under this configuration" to be well supported.

B. External Validity

Do these findings generalize beyond the tested configuration? This is the weakest aspect of the present study. All results use a single population size for the main study ($N = 150$), a single MAGR steering factor and momentum coefficient (both carried over from Vel-v4.3's 2D optimum rather than independently re-derived in 3D), a single scan radius, and a single grid size and trial length. The preliminary $N = 300$ comparison (Section VI) suggests the qualitative gust-response pattern is not an artifact of the specific $N = 150$ configuration, but this was tested at reduced sample size and only three gust levels. We explicitly do not claim that the reported optimum ($G_{\text{wind}} = 4$) generalizes to other population sizes, steering factors, momentum coefficients, scan radii, or grid dimensions without further testing.

C. Construct Validity

Does spatio-temporal efficiency η actually measure what a deployment engineer cares about? η is normalized by both elapsed time and agent count, and is therefore a reasonable proxy for the economic cost of a coverage mission (fewer agent-timesteps per unit coverage is cheaper to operate), but it does not directly capture other properties a physical deployment may require, such as worst-case time-to-completion, energy consumption per agent, or resilience to individual agent failure. No claim in this paper should be read as a claim about those other properties.

D. Summary of Untested Claims

Consolidating the above, we state explicitly what remains untested:

- **No formal convergence guarantee.** Unlike a continuous potential-field formulation, no Lyapunov-style or other formal proof of asymptotic coverage is provided for the discrete nearest-cell steering law studied here. The empirical results in Section VI should be read as evidence of practical behavior under the tested conditions, not as a proven guarantee.
- **No baseline comparison.** This study characterizes Vel-v5.5's own response to wind gust intensity; it does not compare Vel-v5.5 against classical artificial potential field or frontier-exploration methods run under identical 3D, wind-perturbed conditions. Such a baseline comparison is necessary to substantiate any claim of relative advantage over existing methods and is identified as the immediate next step for this line of work.
- **Population size only preliminarily explored.** The main study uses $N = 150$. Section VI reports a preliminary $N = 300$ comparison (20 trials per level, versus 500 for the main study) that finds a consistent efficiency reduction ratio and a preserved qualitative gust-response shape, suggestive of the same scan-overlap congestion Vel-v4.3 documented in 2D. This finding should be treated as preliminary; a full $N = 300$ sweep at matching sample size (500 trials per level), and ideally additional population points as in Vel-v4.3's own three-point sweep, is needed before a population-dependent claim can be made with the same confidence as the gust-intensity findings.
- **MAGR steering factor, momentum, and scan radius not re-swept in 3D.** $G = 1.5$, $\mu = 0.99$, and $r = 5$ are carried over from Vel-v4.3's 2D optimum rather than independently re-derived for the 3D voxel environment. Section III's complexity analysis indicates scan radius in particular may not transfer directly given the cubic rather than quadratic cost scaling in 3D.
- **Simulation only.** No physical hardware validation has been performed. The MVR specifications in Section VII are design targets, not validated hardware requirements.

X. Reproducibility and Data Availability

All 2,500 trial records underlying Table I and the 20-trial-per-level $N = 300$ comparison underlying Table III were generated by a single-file Python simulation implementing Algorithm 1 exactly as specified. Each trial uses an independently-drawn random seed from a common seed sequence, and raw per-trial efficiency values (not only summary statistics) are retained in the accompanying dataset, permitting independent recomputation of every statistic and figure reported in this paper, including the pairwise significance tests in Section VI-D.

XI. Conclusion

Vel-v5.5 extends the empirically-validated Vel-v4.3 MAGR frontier-steering protocol from a 2D cell grid into a 3D voxel environment subject to aerodynamic disturbance. Across 2,500 trials sweeping wind gust intensity, we find that stochastic wind perturbation is not merely a disturbance to be rejected but functions as an exploration-enhancing mechanism: zero-gust operation collapses efficiency to less than half of any wind-perturbed condition tested, efficiency peaks at moderate gust intensity ($G_{\text{wind}} = 4$), and efficiency declines modestly under higher-intensity gust. These findings are reported strictly as empirical observations under the tested conditions; Section IX identifies baseline comparison, population re-sweeping, and formal theoretical characterization of the discrete steering law as necessary next steps before broader claims of protocol superiority can be substantiated.

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